Notes: Andersen, Dimmock, Nielsen, & Peijnenburg (JFE, 2026) Extrapolators and Contrarians: Forecast Bias and Individual Investor Stock Trading

Contents

1. Big picture

- **Question.** Does a person's forecast bias affect how they choose individual stocks to buy and sell, and how they move money into or out of stocks?
- **Main idea.** Extrapolators over-weight recent realizations when forecasting; contrarians under-weight or reverse recent realizations. The paper asks whether that bias maps into real stock-selection decisions.
- **Data innovation.** The authors combine a laboratory experiment measuring individual forecast bias with administrative Danish trading, holdings, income, wealth, and demographic records.
- **Forecast-bias elicitation.** Subjects forecast a stochastic AR(1) process over 40 rounds. Because the true data-generating process is known, the authors can estimate each subject's systematic forecast error.
- **Trading sample.** The laboratory subjects are linked to register data on stock trades from 2011-2021. The design studies purchases, sales, and monthly net flows into stocks.
- **Purchase result.** Investors with higher forecast bias buy stocks with higher past returns. A one-standard-deviation increase in forecast bias is associated with buying stocks with 3.0 percentage points higher prior-year excess returns.
- **Sales result.** Investors with higher forecast bias sell stocks with lower capital gains. For sales, capital gains are more salient than lagged returns because investors can directly see own position-level gains in brokerage accounts.
- **Net-flow result.** Extrapolators increase net flows to stocks after high lagged market returns. Their own portfolio excess returns do not matter in the same way.
- **External-validity result.** Across all Danish investors, investors who make extrapolation-like purchases also make extrapolation-like sales. This supports forecast bias as a common mechanism behind both decisions.
- **Unifying mechanism.** Forecast bias links three performance measures to three choices: past stock returns to purchases, capital gains to sales, and past market returns to net flows.
- **Economic takeaway.** Heterogeneity in how investors process past performance helps explain why different individuals buy different stocks, sell different winners/losers, and change stock exposure differently after market moves.

2. Data and measurement

2.1. Laboratory experiment

- The experiment is designed to measure how individuals process information, not what private information they possess.
- Each subject observes an artificial investment-value sequence and then makes repeated forecasts.
- All subjects face the same kind of task, and the researcher knows the true data-generating process.
- The task follows Afrouzi et al. (2023), but the subjects are a representative sample of Danish investors rather than a convenience sample.
- Subjects are incentivized: a randomly selected forecast can generate a monetary prize, and the winning probability depends on forecast accuracy.

Interpretation: The lab creates a clean measure of forecast bias because subjects see a controlled signal and the true rational forecast is known. This avoids inferring bias indirectly from portfolio choices.

2.2. Forecast-bias variable

- The main variable is ForecastBias_i, the subject-specific slope relating forecast errors to recent realizations.
- $\text{ForecastBias}_i > 0$ means extrapolation: forecasts are too high after high realizations and too low after low realizations.
- $\text{ForecastBias}_i < 0$ means contrarian bias: forecasts are too low after high realizations and too high after low realizations.
- The authors standardize forecast bias so coefficients can be interpreted for a one-standard-deviation change.
- Alternative measures include residualized forecast bias, limited-information forecast bias, rank-based forecast bias, and model-specific diagnostic/sticky/extrapolative/adaptive expectation measures.

Interpretation: The key variable is a stable individual trait in information processing. It is not a survey expectation about the current market, and it is not measured from the same trades used as outcomes.

2.3. Subject recruitment and administrative data

- The subject pool starts from Danish administrative registers, allowing the authors to recruit based on age, homeownership, geography, stockholding, and other variables.
- 959 subjects participate in the experiment.
- The administrative records contain demographics, wealth, income, stock holdings, and stock trades.
- The sample is linked to Refinitiv and Compustat Global returns using ISIN and GVKEY-ISIN mappings.
- Benchmark market returns use the WRDS World Index for Denmark.

Interpretation: The paper's strength is the integration of internally valid lab measurement with externally valid administrative trading data.

2.4. Trading variables

- Purchases and sales are observed from administrative data.
- Lagged annual return is the stock's return over the prior year ending the day before the trade.
- Lagged annual market return is the value-weighted Danish stock-market return over the prior year.
- Lagged annual excess return is stock return minus market return.
- Capital gain is the percentage change in value of an investor's position relative to purchase price.
- Net flows are monthly purchases minus sales, divided by beginning-of-month stock holdings.
- Conditional net flows are net flows conditional on trading that month.

Interpretation: The paper deliberately matches the salient performance measure to the decision: stock returns for purchases, personal capital gains for sales, and market returns for allocation flows.

2.5. Consideration sets

- For purchases, the investor could in principle buy many stocks, but actual attention is limited.
- The authors proxy the consideration set using stocks traded by sample subjects and stocks purchased by at least one Danish investor in the month.

- For sales, the consideration set is cleaner: stocks already held by the investor at the end of the prior day.
- Regressions are weighted so each investor contributes equally rather than letting very active traders dominate.

Interpretation: Purchase tests require a constructed choice set; sales tests are naturally within-portfolio. Investor-day fixed effects then compare stocks available to the same investor on the same day.

3. Models and empirical specifications

3.1. Forecasting task data-generating process

The artificial investment value follows an AR(1) process:

$$x_{t+1} = 100 + 0.5(x_t - 100) + \varepsilon_t. \tag{1}$$

Interpretation: The rational forecast is known because the researcher knows the process. This is what allows the authors to identify systematic forecast errors.

3.2. Forecast-bias estimation

For each subject i, forecast bias is estimated from:

$$F_{i,t}(x_{i,t+1}) - E_{i,t}(x_{i,t+1}) = a_i + b_i(x_{i,t} - \bar{x}) + \varepsilon_{i,t}. \tag{2}$$

- $F_{i,t}(x_{i,t+1})$ is the subject's forecast.
- $E_{i,t}(x_{i,t+1})$ is the rational forecast under the true AR(1) process.
- b_i is forecast bias.
- Positive b_i means extrapolation; negative b_i means contrarian bias.

Interpretation: The slope measures how the forecast error changes with the most recent realization. It directly captures overreaction or underreaction to recent information.

3.3. Purchase specification

The main purchase regression is:

$$\text{Purchase}_{i,j,t} = \beta_1 \text{ForecastBias}_i \times \text{Perform}_{j,t} + \beta_2 \text{Perform}_{j,t} + \delta X_{i,j,t} + \theta_{i,t} + \varepsilon_{i,j,t}. \tag{3}$$

- $\text{Purchase}_{i,j,t} = 100$ if investor i buys stock j on day t, and 0 otherwise.
- $\text{Perform}_{j,t}$ is mainly lagged annual return.
- $\theta_{i,t}$ is an investor-day fixed effect.
- Controls include whether the stock was previously held, current holding, portfolio weight, and prior-month stock purchase popularity.

Interpretation: The investor-day fixed effect makes the test within-person and within-day. It asks whether two investors with different forecast biases choose different stocks out of the same kind of stock universe.

3.4. Purchases by extrapolators and contrarians

The split-sample purchase regression is:

$$\text{Purchase}_{i,j,t} = \beta_1 \text{LaggedAnnualReturn}_{j,t} + \delta X_{i,j,t} + \theta_{i,t} + \varepsilon_{i,j,t}. \tag{4}$$

Interpretation: Estimating this separately for extrapolators and contrarians shows whether the same past-return signal has different effects depending on forecast type.

3.5. Sales specification

For sales, the main regression is:

$$\text{Sale}_{i,j,t} = \beta_1 \text{ForecastBias}_i \times \text{Perform}_{i,j,t} + \beta_2 \text{Perform}_{i,j,t} + \delta X_{i,j,t} + \theta_{i,t} + \theta_l + \varepsilon_{i,j,t}. \tag{5}$$

- $\text{Sale}_{i,j,t} = 100$ if investor i sells stock j on day t.
- $\text{Perform}_{i,j,t}$ is either lagged annual return or capital gain since purchase.
- θ_l is a holding-length fixed effect.
- Controls include portfolio weight and stock-level sales popularity.

Interpretation: The sales regression compares stocks already inside the investor's portfolio. The key prediction is negative when performance is capital gain: extrapolators are more willing to sell lower-gain positions.

3.6. Monotonic forecast-bias bins

The authors divide forecast bias into five bins and estimate:

$$\text{Purchase}_{i,j,t} = \sum_{b=1}^{5} \beta_b I_{i,b} \times \text{LaggedAnnualReturn}_{j,t} + \delta X_{i,j,t} + \theta_{i,t} + \delta_{j,t} + \varepsilon_{i,j,t}, \tag{6}$$

$$\text{Sale}_{i,j,t} = \sum_{b=1}^{5} \beta_b I_{i,b} \times \text{CapitalGain}_{i,j,t} + \delta X_{i,j,t} + \theta_{i,t} + \delta_{j,t} + \theta_l + \varepsilon_{i,j,t}. \tag{7}$$

Interpretation: The monotonic-bin exercise raises the bar for omitted-variable explanations. An omitted factor would need to reproduce a smooth increase for purchases and a smooth decrease for sales across forecast-bias bins.

3.7. Net-flow specification

Monthly net flows are estimated from:

$$\begin{aligned} \text{NetFlows}_{i,t} = {} & \beta_1 \text{MarketRet}_t \times \text{ForecastBias}_i + \beta_2 \text{ExcessRet}_{i,t} \times \text{ForecastBias}_i \\ & + \beta_3 \text{ExcessRet}_{i,t} + \delta X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}. \end{aligned} \tag{8}$$

Interpretation: This shifts from stock selection to asset-class timing. It asks whether extrapolators increase stock exposure after strong market returns.

4. Identification and empirical design

4.1. What identifies the purchase and sales results

- The key variation is the interaction between individual forecast bias and stock-specific past performance.
- Investor-day fixed effects absorb market-wide performance, investor wealth, risk preferences, past experiences, and any other investor characteristic on that day.
- For purchases, the comparison is among stocks in the proxy consideration set.
- For sales, the comparison is among stocks held in the investor's own portfolio.
- The baseline effect comes from how forecast bias changes the investor's reaction to cross-sectional performance differences.

Interpretation: The design separates stock selection from market timing. It asks which stock is selected conditional on trading on that day.

4.2. Why the lab measure helps

- Forecast bias is measured before using the same variable to explain trading decisions.
- The laboratory environment controls the information available to subjects.
- The true rational forecast is known.
- The measure is not derived from observed trading, reducing mechanical correlation.

Interpretation: This gives the study a clean behavioral input: individual differences in information processing are measured independently of the outcome data.

4.3. Potential threats and how the paper addresses them

- **Omitted variables:** the paper adds interactions of past performance with education, financial literacy, numeracy, trust, optimism, and overconfidence, and uses Oster-style coefficient stability.
- **Measurement error:** the paper uses odd/even forecast splits and ORIV logic to show attenuation is not driving the findings.
- **Reverse causality:** the paper repeats the main tests using only trades after the experiment.
- **Choice-set concerns:** the paper tests alternative purchase consideration sets in the online appendix.
- **External validity:** it documents a consistent purchase-sale pattern for the full Danish investor population.

Interpretation: The strongest evidence is the alignment across several margins: purchases, sales, monotonic bins, post-experiment trades, and full-population consistency.

4.4. What the paper cannot fully prove

- The lab task is a generic AR(1) process, not a live stock-forecasting task.
- The true consideration set for purchases is unobserved.
- The analysis is positive, not normative: it does not claim extrapolators or contrarians use an optimal or suboptimal trading strategy.
- The paper cannot observe all information investors use before trading.

- Administrative data provide excellent trading records, but they do not reveal attention, advice, or conversations.

Interpretation: The paper's claim is not that forecast bias is the only driver of stock trading. The claim is that forecast bias is a powerful and measurable mechanism that consistently shapes trading choices.

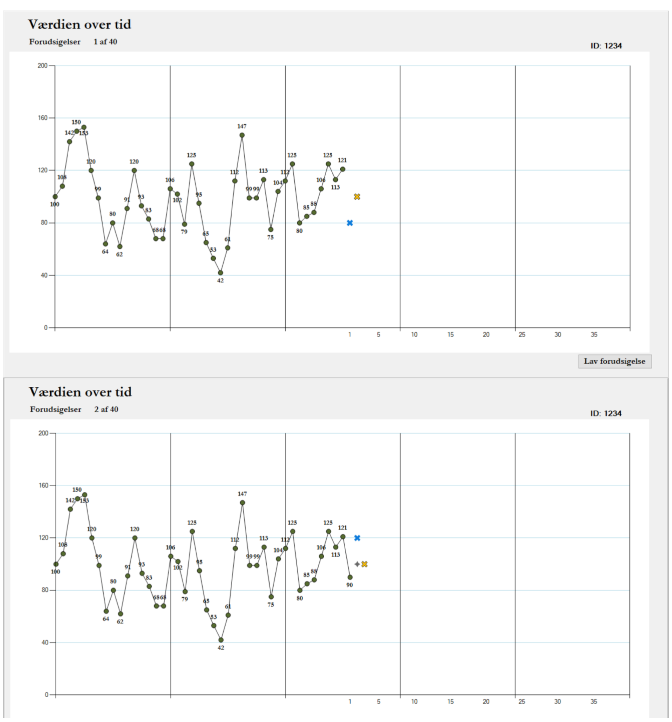
5. Tables and figures

5.1. Figure 1 and Figure 2: Eliciting and measuring forecast bias

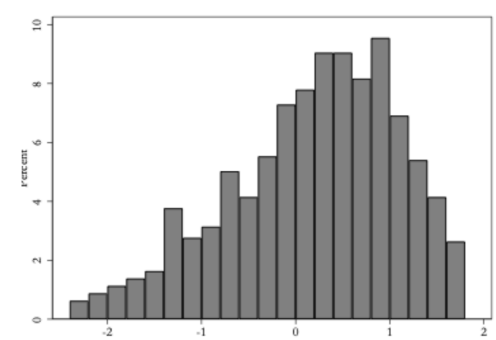
Read:

- Figure 1 shows the forecasting task. Subjects observe past values and choose one- and two-step-ahead forecasts by moving markers.
- After each forecast, the next value is revealed and the subject forecasts again.
- Figure 2 shows the distribution of forecast bias. The mean is positive, so the average subject is mildly extrapolative.
- The distribution contains both extrapolators and contrarians, which creates the cross-sectional heterogeneity used in the paper.

Economic message: The experiment isolates how people process the same type of time-series information. The histogram shows there is enough dispersion in forecast bias to explain heterogeneous trading.



(a) Fig. 1: forecasting task screenshot



(b) Fig. 2: distribution of forecast bias

5.2. Table A1 and Table 1: Variable definitions and summary statistics

Read:

- Table A1 defines the core variables: forecast bias, lagged returns, capital gain, net flows, and demographic controls.
- Table 1 reports trading and individual summary statistics.
- The average investor in the experimental sample has 43.57 buys and 33.10 sales over the sample.
- Average purchase value is 59,299 DKK and average sale value is 82,819 DKK.
- The mean lagged annual return of buy-sample stocks is 26.20%, and the mean capital gain in the sales sample is 23.09%.
- The average subject is 49.7 years old, 72% male, and has 16.47 years of education.

Economic message: The sample contains meaningful real trading activity and substantial cross-sectional heterogeneity in investor characteristics.

Table A1
Variable definitions.

Variable name	Definition
Forecast Bias	The forecast bias parameter estimated as in Eq. (2). For ease of interpretation, we divide the parameter estimate by its standard deviation.
Lagged annual return	The return on the purchased stock over the prior year ending the day before purchase
Lagged annual market return	The return on the value-weighted Danish stock market over the prior year ending the day before purchase
Lagged annual excess return	The difference between the lagged annual return of the stock and the lagged annual market return
Capital gain	The percentage change in the value of the position relative to its purchase price
Held before	Indicator if the subject held the stock previously
Current holding	Indicator if the subject holds the stock currently
Portfolio weight	A stock's weight in the portfolio
Stocks purchase fraction	Prior month purchases of a stock as a fraction of prior month total stock sales in Denmark (multiplied by 100)
Stocks sales fraction	Prior month sales of stock as a fraction of prior month total stock sales in Denmark (multiplied by 100)
Net flows	Value of stock purchases in a month less the value of stock sales divided by the beginning-of-month value of stocks owned (multiplied by 100)
Conditional net flows	Net flows conditional on trading that month
Trade month	Indicator variable equal to 100 in months in which the absolute value of the investor's net flow into stocks is greater than 1% of the beginning-of-month portfolio value
Buy month	Indicator variable equal to 100 in months in which the value of the investor's net flow into stocks is greater than 1% of the beginning-of-month portfolio value
Sell month	Indicator variable equal to 100 in months in which the value of the investor's net flow into stocks is less than -1% of the beginning-of-month portfolio value
Income	The sum of labor income, social transfers, pension income, income from investments, and other personal income, reported in Danish kroner (DKK)
Financial assets	The sum of stocks, bonds, and bank deposits (DKK)
Housing assets	The value of the subjects' home (DKK)
Age	Age in years
Education	Years of formal education
Male	Indicator for male
Married	Indicator if subject is currently married
Children	Indicator for whether the subject has children
Risk aversion	Fraction of paired lottery choice questions for which the subject chose the safer option (details in Online Appendix B)
Financial literacy	Number of the four financial literacy questions answered correctly (details in Online Appendix B)
Numeracy	Number of the three numeracy questions answered correctly
Optimism	Subjects' stated life expectancy less objective life expectancy from actuarial tables (details in Online Appendix B)
Overconfidence	The sum of financial literacy and numeracy questions the subject believes they answered correctly less the number they actually answered correctly (details in Online Appendix B)
Trust	Likert scale where zero indicates "Most people can be trusted" and six indicate "One has to be very careful with other people" (details in Online Appendix B)
Post-Covid experiment	Indicator for subjects whose experimental session was in November 2020

Table A1: Variable definitions

Table 1

Summary statistics. This table reports summary statistics. Appendix Table A1 defines variables. For each variable, we report the mean and standard deviation. The column "quasi-median" reports the average value of the variable for subjects between the 45th and 55th percentile (this is done because our data agreement prohibits reporting non-aggregated values). Panels A and B report summary statistics for the trading and stock characteristics, and individual characteristics, respectively. The summary statistics for the individual characteristics are for the year the experiment is conducted, 2020.

Panel A: Trading and stock characteristics

	Mean	Std. Dev.	Quasi-median
Number of buys	43.57	99.85	14.18
Value of buy	59,299	134,950	25,318
Lagged annual ret. (buys)	26.20%	84.57%	8.86%
Lagged annual excess ret. (buys)	6.41%	82.85%	-9.52%
Number of sales	33.10	75.92	9.94
Value of sale	82,819	624,121	32,815
Prior annual ret. (sales)	29.71%	81.31%	13.74%
Capital gain	23.09%	99.46%	3.37%
Held before (%)	1.79%	13.27%	0%
Current holding (%)	0.21%	4.53%	0%
Portfolio weight (buys)	0.02%	0.90%	0%
Portfolio weight (sales)	8.67%	13.32%	4.14%
Stock purchase fraction (%)	0.06%	0.41%	0.00%
Stock sales fraction (%)	1.07%	1.83%	0.27%
Net flows	0.28	10.63	0
Trade month	13.82%	34.51%	0%
Buy month	7.73%	26.70%	0%
Sell month	6.09%	23.91%	0%
Conditional net flows	2.15	28.37	2.14

Panel B: Individual characteristics

	Mean	Std. dev.	Quasi-median
Age	49.70	7.80	51.16
Male	0.72	0.45	1
Married	0.66	0.47	1
Children	0.81	0.39	1
Education	16.47	2.18	17.04
Financial assets (000's)	2403.89	16,894.96	744.07
Income (000's)	766.37	575.70	644.66
Housing assets (000's)	1916.14	1688.93	1627.67
Post-Covid experiment	0.33	0.47	0
Risk aversion	0.50	0.16	0.49
Financial literacy	3.42	0.79	4
Numeracy	2.84	0.41	3
Optimism	4.65	7.88	5
Overconfidence	0.21	0.88	0
Trust	4.24	1.54	5

Table 1: Summary statistics

5.3. Table 2 and Table 3: Forecast bias and purchases

Read:

- Table 2 shows the main purchase result. The interaction ForecastBias× lagged annual return is positive and significant.
- In the weighted least-squares specification, the coefficient is 0.004 with a standard error of 0.001.
- In the conditional logit specification, the interaction coefficient is 0.061 with a standard error of 0.026.
- Table 3 partitions investors into extrapolators and contrarians.
- The coefficient on lagged return is less negative for extrapolators (-0.004) than for contrarians (-0.013).
- This means that extrapolators tilt purchases toward higher-past-return stocks relative to contrarians.

Economic message: The purchase evidence says forecast bias affects cross-sectional stock selection, not just broad stock-market participation.

Table 2

Forecast bias and stock purchases. This table reports results of regressions of the relation between stock purchases and *Forecast Bias*. Column (1) reports the coefficients of a weighted-least squares regression in which the dependent variable equals 100 if the stock is purchased and zero otherwise, and includes an investor-day fixed effect. Column (2) reports the coefficients of a conditional logit regression in which the dependent variable equals 1 if the stock is purchased and zero otherwise, and conditions out investor-day effects. The key independent variable is *Forecast Bias x Lagged annual return*. The return is winsorized at the 1st and 99th percentiles. The unit of observation is investor-stock-day over the period 2011–2021. In both columns, the observations are weighted such that each investor has equal weight in the regressions. *Forecast Bias* is adjusted to have a standard deviation of one. Both columns include an indicator for a stock previously held by the subject, an indicator for a stock currently owned by the subject, the portfolio weight of a current holding (zero for stocks not currently held), and the fraction of all investor-stock purchases for the full Danish population in the past month that were in the stock. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Weighted-least squares (1)	Conditional logit (2)
Forecast Bias × Lagged annual return	0.004***	0.061**
	(0.001)	(0.026)
Lagged annual return	-0.008***	-0.019
	(0.002)	(0.026)
Investor-Day Fixed Effect	Yes	Yes
Control variables	Yes	Yes
N	33,675,400	33,675,400

(a) Table 2: Forecast bias and stock purchases

Table 3

Past performance and stock purchases for extrapolators and contrarians. This table reports results of weighted-least squares regressions of the relation between lagged annual return and purchases. The dependent variable equals 100 if the stock is purchased and zero otherwise, and includes an investor-day fixed effect. Column (1) presents results for the full sample, column (2) presents results for investors with positive *Forecast Bias,* and column (3) presents results for investors with negative *Forecast Bias*. Returns are winsorized at the 1st and 99th percentiles. The unit of observation is investor-stock-day over the period 2011–2021. The observations are weighted such that each investor has equal weight in the regressions. All columns include an indicator for a stock previously held by the subject, an indicator for a stock currently owned by the subject, the portfolio weight of a current holding (zero for stocks not currently held), and the fraction of all investor-stock purchases for the full Danish population in the past month that were in the stock. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Full sample (1)	Extrapolators (2)	Contrarians (3)
Lagged annual return	-0.008***	-0.004**	-0.013***
	(0.002)	(0.002)	(0.003)
Investor-Day Fixed Effect	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
N	33,675,400	19,545,733	14,129,667

(b) Table 3: Past performance and purchases by forecast type

5.4. Table 4 and Table 5: Forecast bias and sales

Read:

- Table 4 shows that the forecast-bias interaction is insignificant when sales performance is lagged annual return.
- When the performance measure is capital gain, ForecastBias× capital gain is negative and significant.
- The weighted least-squares coefficient is -0.893 with a standard error of 0.401.
- Table 5 splits investors into extrapolators and contrarians.
- For extrapolators, capital gain has a negative and significant sales coefficient (-1.445).
- For contrarians, the coefficient is positive but not significant (0.985).

Economic message: The salient measure for selling is not the stock's generic lagged return but the investor's own capital gain. Extrapolators sell low-gain positions more aggressively.

Table 4
Forecast bias and stock sales. This table reports results of regressions of the relation between stock sales and *Forecast Bias*. Columns (1) and (2) report the coefficients of a weighted-least squares regression in which the dependent variable equals 100 if the stock is sold and zero otherwise, and includes an investor-day fixed effect. Column (3) reports the coefficients of a conditional logit regression in which the dependent variable equals 1 if the stock is sold and zero otherwise, and conditions out investor-day effects. The key independent variables are *Forecast Bias x Performance measure*, where the performance measure is *Lagged annual return* in column (1) and *Capital gain* in columns (2) and (3). Returns and capital gains are winsorized at the 1st and 99th percentiles. The unit of observation is investor-stock-day over the period 2013–2021. In both columns, the observations are weighted such that each investor has equal weight in the regressions. *Forecast Bias* is adjusted to have a standard deviation of one. All columns include monthly holding length fixed effects, as well as controls for the portfolio weight of a current holding, and the fraction of all investor-stock sales for the full Danish population in the past month that were in the stock. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Weighted-least squares		Conditional logit
Performance measure	Lag. annual return	Capital gain	Capital gain
	(1)	(2)	(3)
Forecast Bias × Perform.	-0.414	-0.893**	-0.063**
	(0.674)	(0.401)	(0.027)
Performance measure	3.614***	-0.427	-0.012
	(0.855)	(0.448)	(0.026)
Investor-Day Fixed Effect	Yes	Yes	Yes
Length of Holding Fixed Effect	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

(a) Table 4: Forecast bias and stock sales

Table 5
Past performance and stock sales for extrapolators and contrarians. This table reports results of weighted-least squares regressions of the relation between capital gain and sales. The dependent variable equals 100 if the stock is sold and zero otherwise, and includes an investor-day fixed effect. Column (1) presents results for the full sample, column (2) presents results for investors with positive *Forecast Bias,* and column (3) presents results for investors with negative *Forecast Bias*. Capital gains are winsorized at the 1st and 99th percentiles. The unit of observation is investor-stock-day over the period 2013–2021. The observations are weighted such that each investor has equal weight in the regressions. All columns include monthly holding length fixed effects, as well as controls for the portfolio weight of a current holding, and the fraction of all investor-stock sales for the full Danish population in the past month that were in the stock. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Full sample (1)	Extrapolators (2)	Contrarians (3)
Capital gain	-0.579	-1.445**	0.985
	(0.458)	(0.597)	(0.618)
Investor-Day Fixed Effect	Yes	Yes	Yes
Length of Holding Fixed Effect	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
N	176,451	120,766	55,685

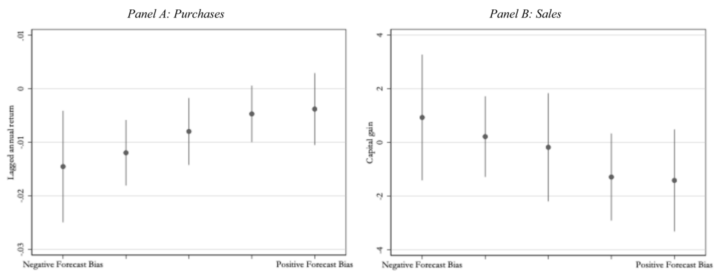
(b) Table 5: Past performance and sales by forecast type

5.5. Figure 3 and Table 6: Monotonicity and post-experiment robustness

Read:

- Figure 3 bins investors by forecast bias and plots the performance sensitivity of purchases and sales.
- For purchases, the coefficient increases as forecast bias rises.
- For sales, the coefficient decreases as forecast bias rises.
- This creates a monotonic pattern across forecast-bias bins.
- Table 6 uses only trades after the experiment.
- The purchase interaction remains positive and significant: 0.003 with standard error 0.001.
- The sales interaction remains negative and significant: -1.046 with standard error 0.482.

Economic message: The monotonic pattern makes omitted-variable explanations harder, and the post-experiment result addresses reverse-causality concerns.



3. Relation between *Forecast Bias* and performance of traded stocks. This figure shows results of regressions of the relation between *Forecast Bias* and s

(a) Fig. 3: forecast-bias bins and traded-stock performance

Table 6
Forecast bias and trading post-experiment. This table reports the coefficients of weighted-least squares regressions using only trading observations post-experiment for purchases (column 1) and sales (column 2). The specification for purchases in column (1) is the same as in column (1) of Table 2. The specification for sales in column (2) is the same as in column (2) of Table 4. The performance measure for purchases (column 1) is lagged annual return and the performance measure for sales (column 2) is capital gain. In both columns, the sample includes only trade-days occurring after the subject participated in the laboratory experiment. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Purchases (1)	Sales (2)
Forecast Bias × Performance	0.003**	-1.046**
	(0.001)	(0.482)
Performance measure	-0.007***	-0.396
	(0.002)	(0.587)
Investor-Day Fixed Effect	Yes	Yes
Length of Holding Fixed Effect	No	Yes
Control variables	Yes	Yes
N	12,416,384	72,771

(b) Table 6: post-experiment trading

5.6. Table 7 and Figure 4: Net flows and consistency across buying and selling

Read:

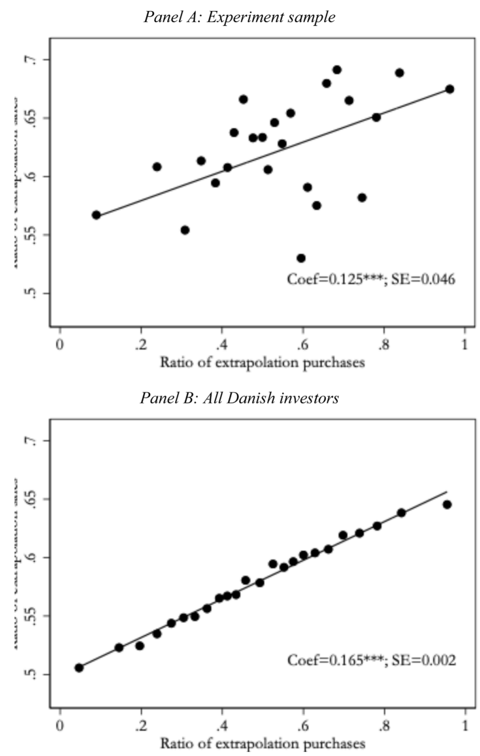
- Table 7 studies monthly net flows into stocks.
- The interaction between forecast bias and lagged market return is positive and significant for net flows.
- The same interaction is also large and significant for conditional net flows.
- The interaction between forecast bias and the investor's own lagged excess return is not significant.
- Figure 4 relates extrapolation-style purchases to extrapolation-style sales.
- Panel A shows consistency in the experiment sample; Panel B shows the same pattern for all Danish investors.

Economic message: Forecast bias affects not only which stocks investors choose but also how they move money into stocks after market-wide performance. The full-population consistency result supports external validity.

Table 7
Net flows, trading, and conditional net flows. This table reports the results of weighted-least squares regressions of monthly net flows, trading activity, and conditional net flows during the period 2012–2021. In columns (1) and (5), the dependent variable is net flows, which is defined as the value of purchases less the value of sales divided by beginning of month portfolio value, and is winsorized at the 1st and 99th percentiles. In columns (2), (3), and (4), the dependent variables are indicators equal to 100 if, respectively, the absolute value of the subject's net flows is greater than 1%, the value of net flows is greater than 1%, and the value of net flows is less than -1%. In columns (1) through (4), the sample includes all investor-months in which the subject owns individual stocks. In column (5), the sample includes only investor-months in which the absolute value of the subject's net flow is greater than 1%. In all columns, the observations are weighted such that each investor has equal weight in the regressions. Lagged annual market return is the return on the Danish stock market index over the prior year. Lagged annual excess return is the subject's stock return over the prior year less the lag market return. All columns include controls for the value of beginning of month stock holdings, financial assets, housing assets, income, education, children, and marital status, as well as individual and year-month fixed effects. Appendix Table A1 defines the variables. Standard errors are clustered at the individual-level and appear in parentheses. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Net flows (1)	Trade month (2)	Buy month (3)	Sell month (4)	Conditional net flows (5)
Forecast Bias × Lag market return	0.854**	-0.739	-0.341	-0.398	6.121***
	(0.346)	(1.057)	(0.782)	(0.705)	(2.151)
Forecast Bias × Lag excess return	0.099	0.547	0.173	0.374	0.457
	(0.194)	(0.594)	(0.432)	(0.409)	(0.900)
Lag excess return	0.080	3.035***	1.656***	1.379***	-0.955
	(0.237)	(0.696)	(0.544)	(0.455)	(1.124)
Investor Fixed Effect	Yes	Yes	Yes	Yes	Yes
Year-Month Fixed Effect	Yes	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes	Yes

(a) Table 7: net flows, trading, and conditional net flows



(b) Fig. 4: consistency between extrapolation purchases and sales

6. Short summary

- The paper asks whether forecast bias affects individual investors' stock-trading choices.
- It elicits forecast bias in a controlled laboratory task and links the measure to Danish administrative trading data.
- Forecast bias is estimated as the slope of forecast errors on recent realizations of a known AR(1) process.
- Positive forecast bias corresponds to extrapolation; negative forecast bias corresponds to contrarian bias.
- Extrapolators buy stocks with higher past returns than contrarians.
- Extrapolators sell stocks with lower capital gains than contrarians.
- Sales are tied more strongly to capital gains than to generic lagged returns because capital gains are personally salient.
- The purchase and sales patterns are monotonic across five forecast-bias bins.

- The main effects survive controls for investor traits, consideration-set changes, measurement-error checks, and post-experiment-only trades.
- Forecast bias also explains heterogeneity in how investors adjust net flows after high market returns.
- In the full Danish population, extrapolation-style purchases and extrapolation-style sales are positively related within investors.
- The paper provides direct evidence that heterogeneity in belief formation shapes real security selection.

7. Conclusion

7.1. Core conclusion

Andersen, Dimmock, Nielsen, and Peijnenburg show that laboratory-measured forecast bias predicts real individual stock-trading behavior. Investors who extrapolate from recent realizations buy stocks with higher past returns, sell positions with lower capital gains, and increase net stock flows after strong market returns.

7.2. Economic intuition

Forecast bias changes how investors interpret salient performance signals. For purchases, the salient signal is the stock's recent return. For sales, the salient signal is the investor's personal capital gain. For net flows, the salient signal is the aggregate market's recent return. The same information-processing bias therefore affects different trading decisions through different performance measures.

7.3. Takeaway

The paper connects experimental finance to household finance. It shows that a bias measured in the lab has predictive power for real administrative trading records, and that forecast bias provides a unifying mechanism behind performance-chasing purchases, capital-gain-sensitive sales, and market-return-sensitive flows.